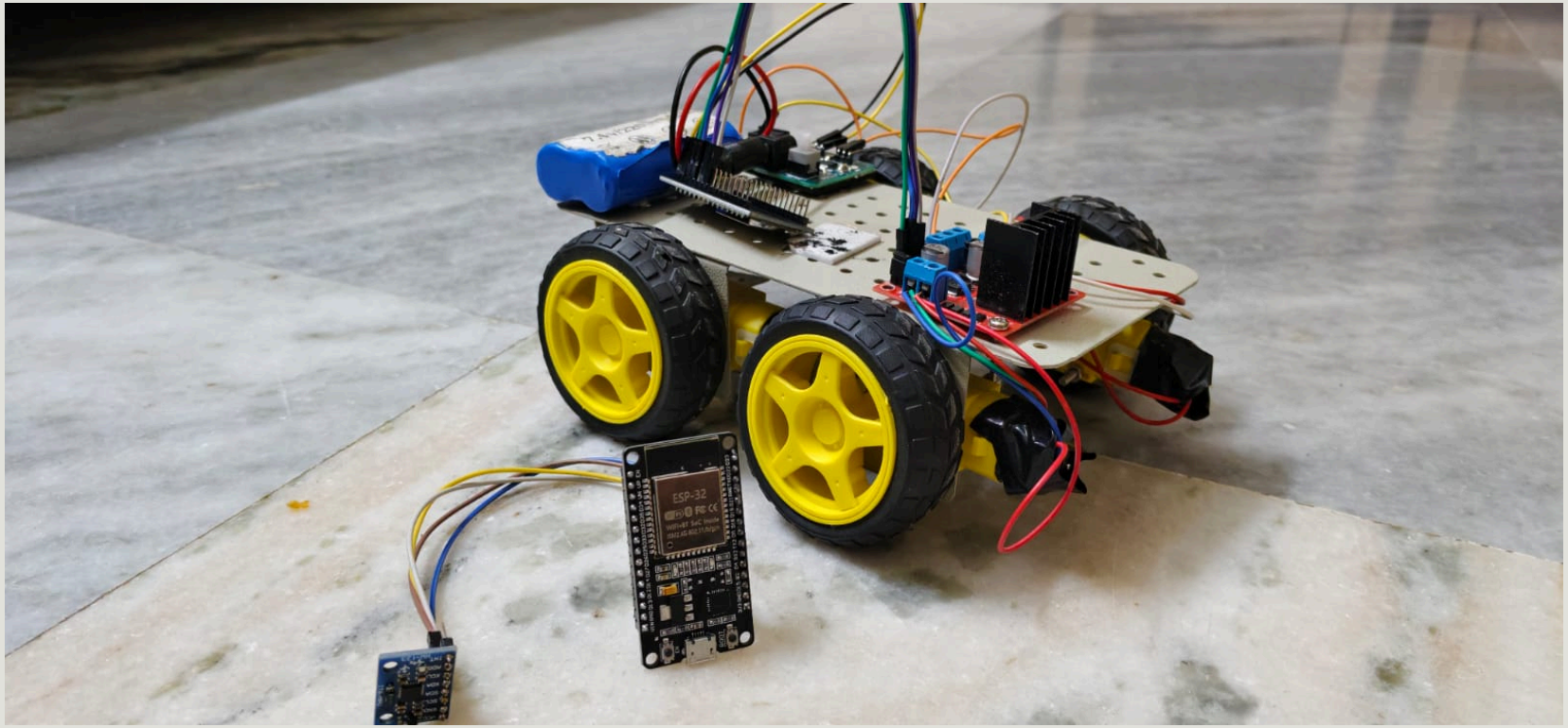
Gesture controlled car



Overview:

Transform hand movements into precise vehicle control using dual ESP32 boards, MPU6050 motion sensing, and wireless communication. This project demonstrates real-time gesture recognition and wireless motor control.

01

Power & Initialization

Battery powers both transmitter (LilyGO T-Display ESP32 + MPU6050) and receiver (ESP32 Module + L298N + BO motors). Systems boot and initialize wireless protocols.

02

Motion Detection

MPU6050 accelerometer and gyroscope capture hand tilt and movement. ESP32 reads sensor data via I2C, processing raw motion values into gesture patterns.

03

Command Generation

Transmitter ESP32 interprets motion data as movement commands: forward, backward, left, right, and stop. Commands are encoded for wireless transmission.

04

Wireless Communication

ESP-NOW transmits commands wirelessly. Receiver ESP32 listens continuously, validates incoming data, and decodes control signals.

05

Motor Control & Movement

Receiver sends PWM signals to L298N driver, which amplifies power for BO motors. Motors rotate wheels based on gesture input, creating real-time responsive movement.

Components:

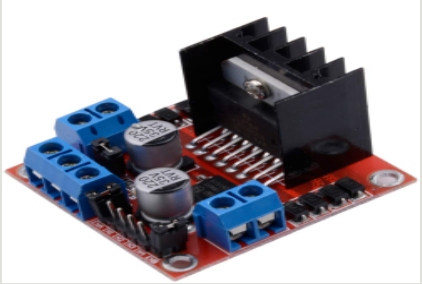
Transmitter:

- ESP32 (LilyGO T-Display)
- MPU6050 Gyro
- 7.4v Li ion battery

Receiver:

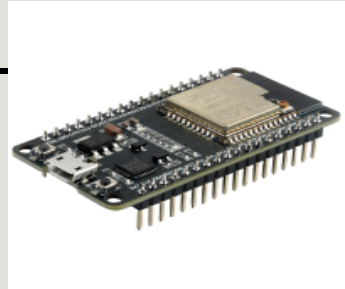
- ESP32 Dev Module
- 2A dual L298N motor driver
- BO motor x4
- 7.4v Li ion battery

- Chassis plates , wheels, jumper wires.



Functions & Configuration

esp32:



Functions:

- Main processing and control unit
- Executes programmed logic and decision making
- Supports wireless communication (Wi-Fi & Bluetooth)
- Interfaces with sensors, actuators, and peripherals
- Handles digital, analog, and PWM signals

Configurations:

- Operating Voltage: 3.3 V
- Processor: Dual-core 32-bit MCU
- Clock Speed: Up to 240 MHz
- Built-in Connectivity: Wi-Fi (802.11 b/g/n), Bluetooth / BLE
- GPIO Pins: Digital I/O, PWM, ADC, UART, I²C, SPI

MPU6050:

Functions:

- Measures acceleration and angular velocity
- Detects orientation, motion, and tilt
- Provides real-time motion sensing data
- Combines accelerometer and gyroscope in one module

Configurations:

- Sensor Type: 6-Axis IMU (3-Axis Accelerometer + 3-Axis Gyroscope)
- Operating Voltage: 3.3 V – 5 V
- Communication Protocol: I²C
- I²C Address: 0x68 (default)
- Measurement Axes: X, Y, Z

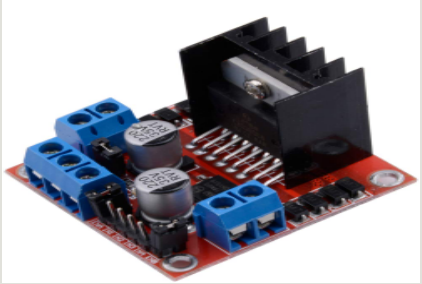
2A Dual Motor Driver (L298N):

Functions:

- Controls speed and direction of DC motors
- Enables forward and reverse motor rotation
- Amplifies low-power control signals
- Interfaces motors with microcontrollers

Configurations:

- Driver Type: Dual H-Bridge
- Motor Channels: 2 DC motors
- Motor Supply Voltage: 7 V – 12 V
- Logic Voltage: 5 V
- Maximum Current: Up to 2 A per channel
- Control Method: Direction pins + PWM



BO Motor:

Functions:

- Converts electrical energy into mechanical motion
- Provides rotation and torque
- Drives wheels in robotic systems
- Acts as the final output (actuator)

Configurations:

- Motor Type: DC Gear Motor (BO Motor)
- Operating Voltage: 3 V – 12 V (commonly 6 V)
- Speed Range: ~100–300 RPM
- Torque: High (gear reduced)
- Rotation: Clockwise and Counter-clockwise

Progress:

Installed Libraries:

- esp_now.h
- WiFi.h
- Wire.h
- Adafruit_MPU6050.h
- Adafruit_Sensor.h

Installed Drivers:

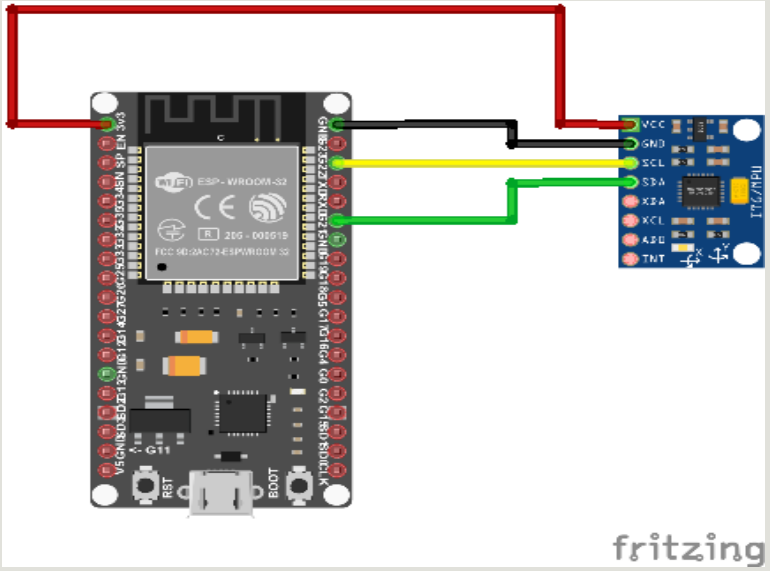
- esp32board
- CH9102X USB Serial Driver
- CP2102 USB Serial Driver

- **Collected the receiver's MAC address (**FC:E8:C0:E1:D0:F0)
- **Programmed the MAC address into the transmitter for communication**

Transmitter Code & Circuit Diagram:

Code → transmitter_mac.ino

Circuit Diagram



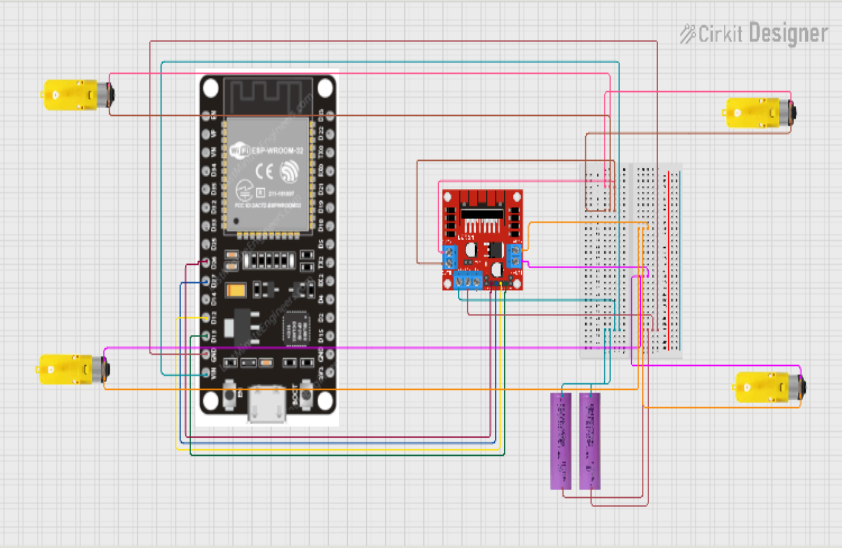
Connections:

MPU6050 / Power	ESP32 Pin
VCC	3.3V
GND	GND
SDA	GPIO 21
SCL	GPIO 22
+ve (Power)	VIN
–ve (Power)	GND

Receiver Code & Circuit Diagram:

Code →

receiver_wl.ino



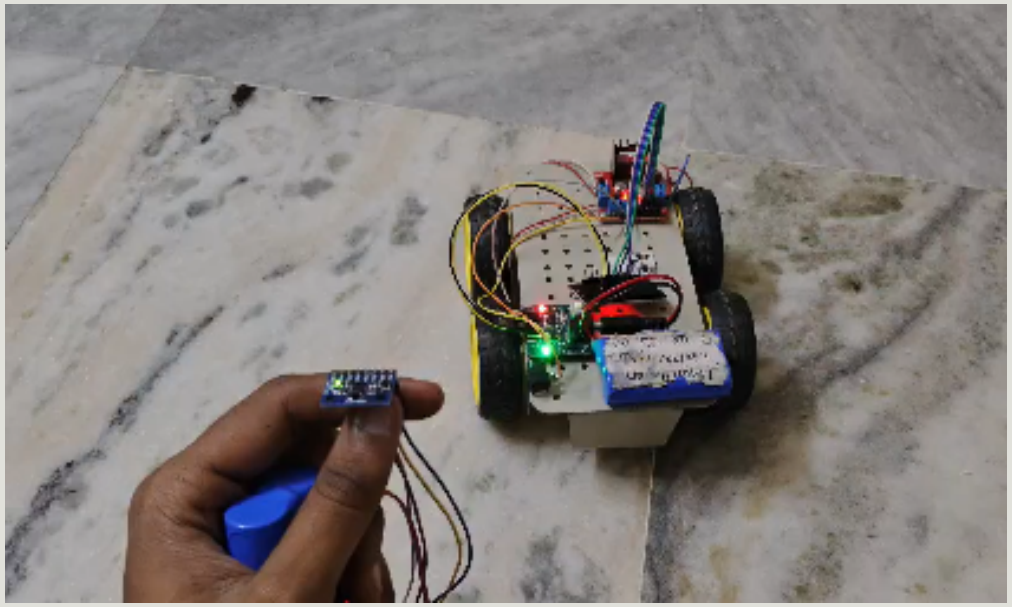
Circuit Diagram

Connections:

Motor Driver / Power	**ESP32 Pin**	**Function**
IN1	D26 (GPIO 26)	Motor-A direction
IN2	D27 (GPIO 27)	Motor-A direction
IN3	D12 (GPIO 12)	Motor-B direction
IN4	D13 (GPIO 13)	Motor-B direction
+ve (Power)	VIN	Power input
–ve (Power)	GND	Common ground

Input & Output:

Transmitter side:
Input: Tilt data from the MPU6050 accelerometer/ gyroscope
(*your hand movement is the actual physical input*)
Output: wireless command sent through the ESP32 (LilyGO T-Display) module.



Receiver side:
Input: Wireless command received by the ESP32 (LilyGO T-Display) (forward/back/left/right/stop)
Output: Motor control signals sent to the L298N motor driver, which drives the DC motor and moves the car

Range:
Indoor: 25 to 30m
Outdoor: 50 to 100m

THANK YOU